

Recommendation system based on Tamil-English code-mixed text analysis

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Abstract— the cinema industry has always been popular since its inception many years ago and is a preferred pastime of many people. It can be observed that even though online movie applications are popular in multilingual society, English is the preferred language. Naturally, people of other languages mix their native language with English during communications resulting in an abundance of multilingual data called code-mixed data, available in today's world. This research focuses on the movie recommendation system whose primary objective is to make a recommender system through Natural Language Processing (NLP) Tools for Tamil-English Code-mixed (Tanglish) Comments. Our recommendation system will be a filtering scheme whose primary objective is to predict a viewer's rating or preference towards a movie or web series.

Keywords— Language prediction, Translation, Sentiment analysis, NLTK, Encryption, Decryption, Tanglish

I. INTRODUCTION

We all know that people like to watch great movies in their spare time, and movies of this era mostly come with "English subtitles", so people of all languages watch those movies and comment in their languages. In this context, this recommendation system provides a mechanism to assist users by classifying users with similar interests. This will allow users to engage in the comments section to read about the movie content, movie strengths, and weaknesses. Then the user will be able to understand the movie more as they know the contents and details about the movie they chose to watch.

This system has four major components. They are Language Identification, Normalization & Translation, Identification of key features and analyzing the sentiment of the Tamil-English code-mixed comments, recommending movies based on Tamil-English code-mixed comments, and a rating system for movies based on Tamil-English code-mixed comments.

Language prediction is an essential step in Natural Language Processing (NLP). It involves trying to predict the natural language of a piece of text. Before taking other

actions (translation/sentiment analysis), it is essential to know the language of the text. For example, if person visits Google Translate, the text field available to type in says "Detect language." This is because Google first tries to detect the language of the sentence before translating it. Therefore, this activity is considered the first component of this system.

The recommended approach is useful, and the sense structure can contribute to understanding the meaning of short text. It also reveals the influence of different relationships between the modifier and the emotion word, and the contribution of each sentence to the emotion computation of the short text is different. In sentiment analysis, the main problems with current techniques are the inability to perform well in different domains, insufficient accuracy and efficiency in sentiment analysis based on well-labelled data, inability to handle complex sentences requiring more than emotional words, and simple analysis. The major problem is still related to correctly interpreting the context in which certain words are used. It is still difficult for most tools to accurately assess what constitutes a negative, neutral, and positive statement. To solve this problem, word2vec for sentiment analysis achieves high precision and recall rate. Follow-up work will consider extracting the features of the material described in the text through some method such as Linear Discriminant Analysis (LDA) or bootstrapping method. The system analyzes the sentiment of features and conduct more meaningful research predicting sentiment polarity with greater sentiment knowledge. The network model and training method of SINM is not complicated and can achieve better results.

The proposed system contains a movie recommendation system that will provide users with content by considering what other people recommend and users' selections. In the rating system, identifying different cohorts of users/customers, like prediction churn, lifetime value, and product preferences, is risky. Furthermore, it was challenging to accurately identify and extract different types of feedback (positive and negative reviews/comments, mentions of attributes such as clothing size, etc.). In the rating system, several options provide users of an application with a better sense of the relative ranking of movies. A primary function is to show users the average score for all movies in the relevant category(e.g., Action, Romantic, Horror, Cartoon,

etc.) Collaborative Filtering is a widely used solution for this problem used in the system.

A movie recommendation is important in our social life due to its strength in providing enhanced entertainment. There have been several challenges in determining whether a system can effectively suggest a set of movies to users based on their interest or the popularity of the movies in Sri Lankan entertainment.

The proposed application “MOVREC” will help users to find movies of their choice efficiently and effectively without wasting much time on useless browsing based on the movie experience of real movie enthusiasts.

II. LITERATURE REVIEW

Dwijia Parikh et al [2]. (2021) conducted a study about “Normalization and back transliteration for code-switched data”. They aimed to provide a system for transliterating Romanized Hindi to the Devanagari orthography. For that, they used normalization, translation, and transliteration. Also, as techniques, back transliteration and Grapheme-to-Phoneme (G2P) conversion were found. The integrated system showed the correct script text of back-transliteration, and the overall result gives 80.48 marks for BLEU and 18.5 marks for Word Error Ratio (WER) [2]. In the future, this can easily be extended to other languages that employ variations of the Devanagari script, such as Gujarati and Nepali.

In a study conducted by Anab Maulana Barik et al. [1] (2020), it was stated that the normalization of Indonesian-English code-mixed Twitter data from the pipeline is composed of four modules: tokenization, language identification, lexical normalization, and translation. Furthermore, the out-of-vocabulary (OOV) [5], Longest Common Subsequence Ration (LCSR), and Longest Common Subsequence (LCS) techniques [1] were used. The synthesized system has shown that the model works effectively and that overall, the pipeline yields a score of 54.07 for BLEU and 31.89 for Word Error Rate (WER) [1]. The identified drawback is that because it is a pipelined process, the process error from each module propagates to the next module because each model uses the output from previous modules as inputs.

A study was conducted by Vivek Srivastava et al. [3] (2020) about an “English-Hindi Code-Mixed Machine Translation System”, and the name is “Parallel Hinglish Social Media code-mixed corpus (PHINC)”. They aimed to provide a parallel corpus for findable, accessible, interoperable, and reusable English-Hindi code-mixed data. They also used the Annotation technique, Code Mixing Index (CMI), Quality of Translations (QT), Pipeline, and Google Translate as techniques [3]. Their synthesized state-of-the-art machine translation system for monolingual corpus has shown various challenges, and the performance of various systems on their parallel corpus has been demonstrated, and the overall result has yielded 0.814 scores for BLEU1, 0.704 scores for BLEU2, 0.565 scores for BLEU3, and 0.388 scores for BLEU4 [3]. Various code-mixed languages could also be explored using it in the future, especially those that are low resource and endangered.

According to a study conducted by Saini Kumar Mahata et al. [4] (2019), the Normalization of Indonesian-English code-mixed Twitter data from the sequence-to-sequence model, which is the development of a framework for code-mixed to monolingual machine translation without employing any parallel corpus for training. Also, the seq2seq model, the token reordering module and matrix/embedded language were used [4]. The integrated system shows that the model works effectively, and overall, the system achieves a BLEU score of 16.47 [4].

J. Bobadilla et al. [6] have reported that the performance of collaborative filtering is based on deep learning. Their system got 16.47 as a BLEU score, which was a good starting point for their research. Deep learning has extracted the nonlinear relationship between predictions, reliabilities, and correct suggestions. Three abstraction layers—real prediction errors, prediction errors, and projected ratings—help suggested architecture perform better [7].

The authors of sentiment observations [7] addressed SA at the document level. They suggested combining supervised and unsupervised algorithms with rich sentiment content for learning word vectors (also called word description) capturing techniques, which include continuous, multi-dimensional sentiments and non-sentiment annotations. The authors used IMDB to evaluate the movie review dataset, and they found that their model performed better than other learn vector capturing approaches. BLEU received a 19.5 rating.

Gilbert Badaro et al. developed a hybrid technique that uses a weighted mix of user- and item-based filtering to locate undiscovered ratings for an item [9] so that the top-ranked item may be promoted. Item rating is predicted by multiplying the ratings of 'UBCF' and 'IBCF' by the weights assigned to them. The author demonstrated how this combination also aids in overcoming the issues of cold start users and cold start items. scores for the combination strategy of [8] BLEU and 15.5. [10]. The method also takes into account the fact that accuracy and data sparsity are both decreased. The dataset for movie lenses is utilized for the experiment.

F. Deng et al. [9] provided a method to convert user-item ratings into hybrid feature ratings to determine a user's probable preference based on hybrid features such as user-produced features, picture visual features, and others. The experiments [9] demonstrated that the suggested method has superior efficiency on an extensive dataset and produces better results on a sparse dataset. In their paper, C. Yang et al. [8] proposed a hybrid technique based on item characteristics and social similarity. The author integrated social commonalities and movie genres with the collaborative filtering process. To address the sparsity issue, he employed the BPR-MF model. The suggested approach operates in two steps. First, the candidate set is obtained using the ratings from the training dataset using the BPR-MF model.

Akshi Kumar et al. [14]. (2012), published a study on “Sentiment Analysis: A Perspective on its Past, Present and Future”. They aimed to gain perspective on sentiment analysis, expounding its basic terminology, tasks and granularity levels. For that, they used 101 topics, and it had 5,699 articles. As a result, sentiment analysis has multiplied fifty-fold over the past ten years, making it one of the

fastest-growing academic topics in recent memory. Before making any decisions in the early days of the internet, a person might get advice from his friends, neighbours, and family. Organisations conducted surveys, opinion sampling, and broad public opinion polls on their goods or services. The development of the World Wide Web, and in particular the creation and popularity of Web 2.0, which emphasises user-generated content, has substantially altered how a person expresses his or her opinions. People may now express their ideas on their personal websites through blogs, social media sites, forums, and reviews. Big data results from sophisticated methods for handling large-scale, multidimensional data and has drastically improved storage capacity and intricate data formats. Modern technology and strategies are needed in this field to handle many computational challenges and capture usable data without losing sensitive data. To solve these issues, machine learning has lately been presented as a new and quickly growing field of study.

R.Swathi et al [11]. (2017), published a study on “Systematic survey on the evolution of machine learning for big data”. They aimed to categorize recent publications. To that end, 54 recently published articles based on sentiment analysis were selected, classified, and summarized. The investigation of ideas, thoughts, feelings, and subjectivity in a text is referred to as sentimental analysis. Social media serves as a vital resource where people may connect and find the things they need. Customer and business satisfaction are both increased by this. Recent algorithms, emotional analysis methods, and the significance of disciplines like transfer learning, emotion detection, and resource construction are all covered.

Christopher C. Yang et al [13]. (2010) researched “Understanding Online Consumer Review Opinions with Sentiment Analysis using Machine Learning”. Their aim was to user-contributed reviews of products, which many online consumers use to support their purchase decisions by identifying products that best fit their preferences. They proposed two supervised learning techniques (class association rules and naïve Bayes classifier) to classify opinion sentences into appropriate product feature classes and produce a summary of consumer reviews. An empirical evaluation that compares the performance of the class association rules technique and the naïve Bayes classifier for sentiment analysis shows that their proposed techniques achieve more than 70% of the macro and micro F-measures.

III. METHODOLOGY

The proposed solution aims to provide a recommendation system for movie enthusiasts or lovers to recommend a suitable movie for the appropriate genres (action, romance, comedy, horror). Figure 1 shows that movie comments are initially extracted from YouTube using the scrape method and saved as an ‘XLSX’ file. Then comments are imported to our recommendation system. These comments are first subjected to ‘Language identification normalization & translation’. Those comments are translated into Tanglish comments, followed by “Identifying key features and sentiment analyzing”. Here, positive comments and negative comments are identified from Tanglish comments. The film is recommended according to the appropriate genres in the section ‘Recommending movie based on Tamil-English

code-mixed comments’. The methodology followed to implement the recommendation system is discussed in this section.

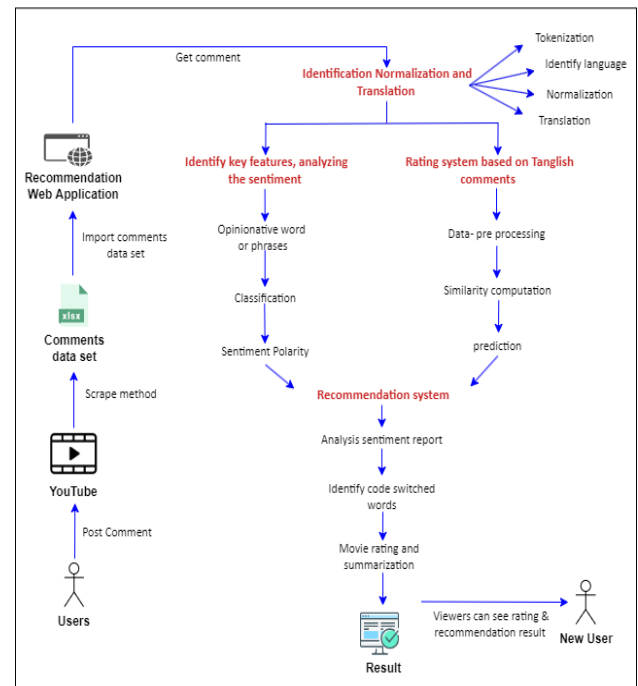


Figure 1: Overall of the diagram of the proposed solution

A. Language Identification normalization & Translation

The first step is to predict whether the language of the comments will be English, Tamil, or code-mixed. This was accomplished using preprocessing techniques. The following step is to translate the comments into English while continuously removing stop words. The English comments must then be translated into Tamil to obtain the Tamil meaning of the comments. Subsequently, the dictionary created for this component encodes and decodes the comments.

To see the process in a detailed view, initially, the extracted movie comments are saved into an ‘excel’ file. The comments must be imported into the language identification, normalization & translation component. The dataset is accessed using the ‘read_excel’ function from pandas’ library since it is an ‘XLSX’ file. Following that, data cleaning, language detection, and tokenization are required. First, comments are cleaned using a method which was created to remove the punctuation marks. Then ‘GOOGLE TRANS’ is used for language prediction and translation process. Here, the comments are translated into English-language. Because it will be easier for the next process if it is in a common language. The next step is to perform a continuous tokenization process as the stop words need to be removed. The balance tokens need to merge after removing the stop words. For this purpose, a method has been created to combine tokens.

As the next step, all the ‘merge-tokens’(English-comments) from the dataset should be translated into Tamil-language. The Tamil meaning of all comments is required to create Tanglish data. Because Tanglish means letters would be in English-language and meaning would be in Tamil-language. Therefore, all comments have been

translated, which were merged in the previous function. The target language is selected as 'Tamil' and the comments are translated with the help of Google Translate. In addition, the system creates dictionaries. Instead of Unicode, new solutions are introduced by creating two separate dictionaries for Tamil and English alphabets. This is because reading past research papers on topics was mostly done with Unicode. Python dictionary creation method is used for creating these dictionaries, as shown in Figure 2. Assuming 'Tamil-alphabet-dictionary', it contains Tamil vowels, consonants, and special characters, and 'English-alphabet-dictionary' content is similar to 'Tamil-alphabets-dictionary'.

```
dictionary_tamil = { "அ" : "@", "இ" : "#", "உ" : "%" }
dictionary_english = { "@": "A", "#": "I", "%": "U" }
```

Figure 2: Dictionary model for encoding & decoding

Tamil comments are encoded using a dictionary developed for Tamil alphabets, as shown in Figure 3. All the Tamil comments are encoded into code-mixed data. For encryption, a new algorithm has been developed using the concept of 'symmetric' and 'asymmetric' under cryptography. After that, by using this algorithm, a method was created for the encoding process.

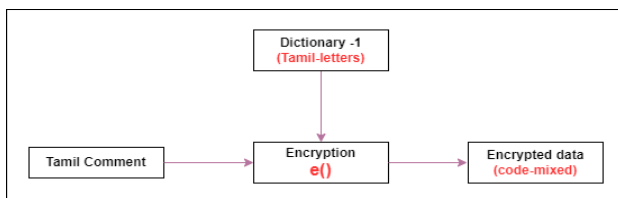


Figure 3: Process of encoding using Tamil-alphabets-dictionary

The data encrypted in the previous step must be decoded using the 'English-alphabets-dictionary' to get the Tanglish comments. For decoding, a new algorithm was developed using the concept of 'symmetric & asymmetric' under cryptography, similar to encryption. Afterwards, by using this algorithm, a method has been created for the decoding process. Then the encrypted data is decoded with the help of 'English-alphabets-dictionary'. Figure 4 shows this process. At the end of this process, Tanglish comments were obtained as output.

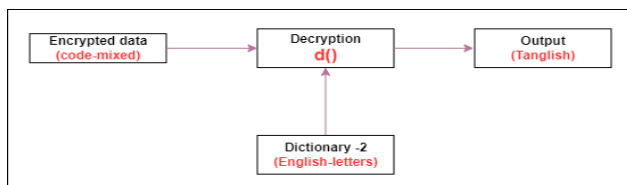


Figure 4: Process of decoding using English- alphabets-dictionary

As summarized in Figure 5, the input imported Comments are translated into English with the NLTK process as a first step. Then they are translated into the Tamil language. Next, the conversion of appropriate symbols for each Tamil character was done through the Tamil-alphabets-dictionary (encoding). After that, the decoding process was done using the second dictionary, which is the English-alphabets-dictionary, and finally, Tanglish data was given as output.

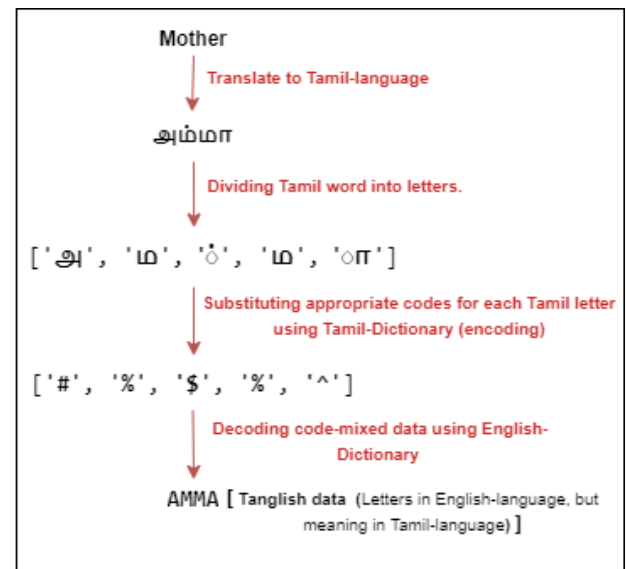


Figure 5: Process of Tanglish translation

Finally, comments have been translated into a single language which is Tanglish. Next, Tanglish comments need to be passed to other processes. So, Tanglish comments are saved as an 'XLSX' file using the 'ExcelWriter' function from the 'pandas' library.

B. Identifying key features and Sentiment analyzing

Using these component makes it possible to detect and understand their feelings from customers comments. Sentiment analysis tools generate insights on how to enhance the customer experience and improve movie suggestions. Thus, the system can understand how customers feel about the movie.

First, it is necessary to import Tanglish comments, which is the output from the 'language identification, normalization & translation'. Tanglish comments have been imported into the system through the 'read_excel' function from 'pandas' library since it is a 'XLSX' file. Every word can be extracted from the Tanglish comments by tokenization. Because it would be helpful when comparing with our list, which has been created for sentimental words, a method was created for tokenization. As shown in Figure 6, two lists have been created for positive and negative sentiment words. Python basic list-creation-method is used for creating these lists. Here positive-list contains all positive words in Tanglish and a negative-list contains all negative words in Tanglish.

```
positive_list = { "sema", "nalla", "arumai", "arumayana", "sattapadi": "super" }
negative_list = { "mokka", "kevalam", "seruppu", "vaapillai", "pidikkavillai" }
```

Figure 6: Positive & Negative word list model

Then compare the tokens of Tanglish comments with the lists to check whether they match it. 'String matching algorithms' have been used for this process, and a method is created using these string algorithms.

After finding the sentimental words from the Tanglish comments, a percentage can be calculated using the number of positive and negative comments for a movie. A basic mathematical function has been used, as shown in equations 1 and 2.

$$PCP = \frac{PCC}{(PCC + NCC)} \times 100 \quad (1)$$

Where PCC and NCC represent the positive and negative comments counted, respectively. The ratio of positive comments to the total number of comments can be calculated by dividing the PCC by the sum of the PCC and NCC. Finally, multiply this value by 100 to get PCP or positive comment percentage. Similarly, the negative comment percentage (NCP) is also computed in (2).

$$NCP = \frac{NCC}{(PCC + NCC)} \times 100 \quad (2)$$

If there are more positive comments percentages rather than a negative percentage, then the movie is marked as good. In contrast, if there are more negative comments percentage rather than a positive percentage, then the movie is considered bad.

c. Recommending movie based on Tamil-English code-mixed comments

Using these component makes it possible to find movie genres from customers comments. Here, it is necessary to import the Tanglish and sentiment analysis report dataset, which are the outputs from the Components one and two that are 'language identification, normalization & translation' and 'identifying key features and sentiment analyzing'. These datasets have been imported into the system through the 'read_excel' function from 'pandas' library.

After that, these data are subjected to a tokenization process. At the same time, four lists have been created for film categories and Figure 7 shows these lists as well. Then compare the tokens with the genres-lists to check whether they match it. 'String matching algorithms' have been used for this process, and a method is created using these string algorithms.

```
comedy_list = ['sirippu', 'siriththan', 'comedy', 'haha', 'lol', 'lollu', 'Piramehan', 'siripputhan', 'funthan',
action_list = ['sandai', 'sandaikal', 'sandaikasikal', 'adipadu', 'neruppu', 'tharathappada', 'paight', 'gunshootah
love_list = ['kathal', 'lovers', 'kathalarkal', 'kathalar', 'kaviyam', 'kathalkaviyam', 'lovesairukku', 'romantic',
horror_list = ['payama', 'payanthudan', 'thikil', 'pei', 'pisasu', 'killer', 'serial', 'pakkupakku', 'tention', 'c
```

Figure 7: Lists model for movie genres

After finding the matching tokens of each movie genre, percentage can be calculated for the genres and Figure 12 represents it. For instance, if 'comedy' genre has a high percentage than other genres that movie will recommend as comedy movie. Sometimes two genres may have same percentage value. That time system will recommend that particular movie for these two genres.

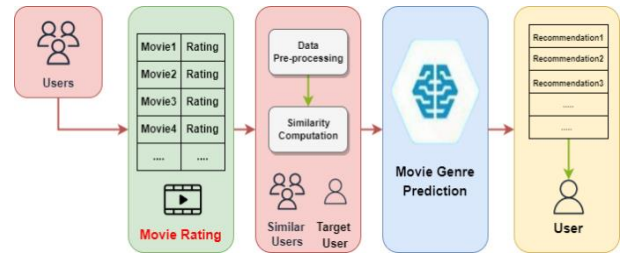


Figure 8: Procedure of recommendation system

As summarized in Figure 8, system has two kind of users such as 'similar' user and 'target' user. These users can add some movies comments to the recommendation system. Then system will analysis that movie comments, which means data pre-processing, and similarity computation process. Afterwards the system takes movie-genre prediction process. Finally, system will show the recommending result to the particular user.

IV. RESULTS AND DISCUSSION

Cinema has become one of our most popular and unavoidable hobbies. However, people like to spend their precious leisure time watching good or best movies, and this research was able to help people achieve that goal successfully.

Language identification normalization and translation function have two outputs. These are 'encoding response' and 'decoding response'. Figure 9 is referring encoding result. This result cannot read by the users. Because it is in as code-mixed data.

```
===== En-Code =====
bo*o+n+pan #o+v#n#ohocn -oanfofo 4hnho+r
2+r -ogngo bo*ocn
-oan #rio=n+vdof# 2er=n+o+pgper=n+r jp^odn eo%p#oen#ohn
2=n+o bo*ocn 6ao#n#r bp*p#n#o+jpgngv
cp#o 0io#oao bo*ocn
```

Figure 9: Encoded comments output of the encoding process

Figure 10 is referring the final output (Tanglish comments) of language identification normalization and translation component. At the end of this process, 'Tanglish' comments had obtained. On perusal of the output, it was observed that the comment letters were in English, and the meaning was in Tamil.

```
===== De-Code(Tamil-English Code-mix Data) =====
Padaththin kathaikkalam nanraka ullathu
ithu nalla padam
nan kulanthaiyaka irunthathilirunthu vijay rasikarkal
intha padam enakku pidikkavillai
mika alakana padam
```

Figure 10: Tanglish comments output of the decoding process

Figure 11 shows the result of identifying key features and analyzing the sentiment of the Tanglish comment component. There are more positive comments than negative comments. Therefore, the system would give positive review for to the movie.

Positive Comments = 80.0%
Negative Comments = 20.0%

Figure 11: Output of positive & negative comments.

Figure 12 shows the movie recommendation output based on Tanglish comments, and the percentage of each category can be seen here. Therefore, the system will recommend this movie to the category with the highest percentage. Sometimes two categories may have the same percentage, and the system recommends both categories. For instance, “This movie is recommended for 'Comedy and Love”.

Action percentage : 12.5%
Comedy percentage : 37.5%
Romance percentage : 12.5%
Horror percentage : 37.5%

Figure 12: Sample output of movie genres.

V. CONCLUSION AND FUTURE WORK

This paper presents a movie recommendation system based on Tamil-English code-mixed data. Various challenges posed by the Tanglish translation method for Tamil-English code-mixed data while dealing with film concepts were discussed. The effectiveness of various system or component has also been demonstrated in this recommendation system.

From this research 90-95% successfully expected result had obtained. But even a new drawback appeared here. That is coming under the first component, which is ‘language identification normalization & translation’. Since, we are using ‘GOOGLE TRANS’ for the language prediction, it takes a long time to predict the language. Therefore, as the number of data sets increases, the processing time increases. This is the only drawback of our research background. In future, this drawback will try to solve by us. Also, by using dictionaries, the encoding and decoding processes work correctly and faster.

From the various user comments, there is an excellent opportunity to develop systems that can overcome the challenges associated with the code-mixed text. This system can be used for other native languages as well. For instance, if take Sri Lanka, system can be changed to ‘Recommendation system based on Sinhala-English Code-mixed data’. Various code-mixed languages can also be explored. The system presented here can also be extended for various user’s comments like a movie trailer, teaser, entertainment TV shows etc.

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